

An alternating antimorphic Fine–Wilf theorem

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Abstract

Let θ be an antimorphic involution of a free monoid. If a finite word w has alternating θ -periods p, q and $|w| \geq p + q - \gcd(p, q)$, then $\gcd(p, q)$ is an alternating θ -period of w . This proves the unnumbered conjecture stated after Theorem 23 in Nowotka’s contribution, joint with Bischoff, to *Mini-Workshop: Combinatorics on Words* (2010). The proof applies the classical Fine–Wilf theorem to the common reflected extension $\theta(w)w$ and recovers the alternating phase at its center. The length formula cannot be decreased by one uniformly.

1 The statement

Let A be an arbitrary alphabet, and let A^* be its free monoid, with empty word ε . An *antimorphic involution* is a map $\theta : A^* \rightarrow A^*$ such that $\theta(xy) = \theta(y)\theta(x)$ and $\theta^2 = \text{id}$. A positive integer p is an *alternating θ -period* of a finite word w if there is a word $u \in A^p$ such that w is a prefix of $(u\theta(u))^\omega$. The periods need not be minimal. This definition prescribes alternating blocks; it differs from a prefix of an arbitrary concatenation of u and $\theta(u)$.

Theorem 1. *Suppose that w has alternating θ -periods p and q , and put $d = \gcd(p, q)$. If $L = |w| \geq p + q - d$, then, for the length- d prefix v of w , the word w is a prefix of $(v\theta(v))^\omega$.*

This is the exact alternating conclusion in the unnumbered conjecture following Theorem 23 on printed page 2220 of [2]; it is distinct from numbered Conjecture 27 there. Theorem 23 gives the sufficient bound $p + q$. Bischoff’s thesis [3, Lemma 3.9 and Satz 3.11] also develops reflection and doubled ordinary periods and proves that longer bound. We use those classical symmetries with a common extension of the entire finite prefix. Fixed letters of θ are allowed; in particular, ordinary reversal is included.

2 The common extension

For a word of length n , an *ordinary period* $h > 0$ means that letters in positions i and $i + h$ agree whenever both positions occur. We use the classical Fine–Wilf theorem [1]: a word with ordinary periods h, k and length at least $h + k - \gcd(h, k)$ has ordinary period $\gcd(h, k)$.

Lemma 1. *There is an involutive permutation $\sigma : A \rightarrow A$ such that*

$$\theta(a_0 \cdots a_{n-1}) = \sigma(a_{n-1}) \cdots \sigma(a_0).$$

Proof. The identity $\theta(\varepsilon) = \theta(\varepsilon)^2$ forces $\theta(\varepsilon) = \varepsilon$. Since θ is bijective, nonempty words have nonempty images. If the image of a letter were xy with both factors nonempty, applying θ would factor that letter as the two nonempty words $\theta(y)\theta(x)$, a contradiction. Thus letters map to letters. Involutivity and the antimorphism rule give the assertion. $\square$

Lemma 2. *If $L = |w| \geq p$ and p is an alternating θ -period of w , then $\theta(w)w$ has ordinary period $2p$.*

Proof. Write $u = w_0 \cdots w_{p-1}$ and extend the block $u\theta(u)$ periodically to a sequence $(s_i)_{i \in \mathbb{Z}}$ of period $2p$, with $s_i = w_i$ for $0 \leq i < L$. If r is the residue of i in $\{0, \dots, 2p-1\}$, then

$$s_i = \begin{cases} u_r, & 0 \leq r < p, \\ \sigma(u_{2p-1-r}), & p \leq r < 2p. \end{cases}$$

The residue of $-1-i$ is $2p-1-r$, so $s_{-1-i} = \sigma(s_i)$ in both cases. Hence, on the negative positions $-L \leq j < 0$, we have $s_j = \sigma(w_{-1-j})$. The restriction to $[-L, L-1]$ is therefore exactly $\theta(w)w$, indexed with w_0 at position 0. As a restriction of a $2p$ -periodic sequence, it has ordinary period $2p$. $\square$

The extension in Lemma 2 is determined by w alone. In particular, applying it with p and q yields ordinary periods on the *same finite word*. No equality of two unseen infinite extensions is being assumed, and no condition on the last incomplete block of w is needed.

Proof of Theorem 1. Since $d \leq \min(p, q)$, the hypothesis gives $L \geq \max(p, q)$. Lemma 2 shows that the word $W = \theta(w)w$, of length $2L$, has ordinary periods $2p$ and $2q$. Moreover,

$$2L \geq 2p + 2q - 2d, \quad \gcd(2p, 2q) = 2d.$$

Fine–Wilf therefore gives ordinary period $2d$ for W . Continue to index W on $[-L, L-1]$. Its central restriction to $[-d, d-1]$ is $\theta(v)v$, where $v = w_0 \cdots w_{d-1}$.

For $0 \leq i < L$, let $r \in \{0, \dots, 2d-1\}$ be the residue of i modulo $2d$. If $r < d$, choose $j = r$. Otherwise choose $j = r - 2d \in [-d, -1]$. In each case j and i are in $[-L, L-1]$ and differ by a multiple of $2d$. All intermediate positions between them remain in that interval, so ordinary period $2d$ gives $W_i = W_j$. Consequently

$$w_i = \begin{cases} v_r, & 0 \leq r < d, \\ \sigma(v_{2d-1-r}), & d \leq r < 2d. \end{cases}$$

These are precisely the letters of $(v\theta(v))^\omega$. $\square$

Proposition 1. *The sufficient formula $p+q-\gcd(p, q)$ cannot be replaced uniformly by $p+q-\gcd(p, q)-1$.*

Proof. Take distinct letters a, b , let θ be reversal, and set $w = abb$, $p = 2$, $q = 3$. The word abb is a prefix of $(abba)^\omega$ and of $(abbba)^\omega$, so both alternating periods occur. Its length is $3 = 2 + 3 - 1 - 1$, but alternating period 1 would force the constant word $aaa \cdots$. It does not occur. $\square$

This example establishes only the stated uniform obstruction. It does not assert a sharp minimum length for each pair (p, q) , alphabet, or involution. Theorem 1 also covers $p = q$ and divisibility of one period by the other.

3 Related results and verification

The ordinary Fine–Wilf threshold and reflection/doubled-period mechanisms are prior work [1, 2, 3]. Theorems 25 and 26 of Czeizler–Kari–Seki [4] concern a θ -palindromic base word and common θ -primitive roots. Their Theorem 28 doubles a base block to $u\theta(u)$; Corollary 32 treats more general block powers at the longer bound $2p + q - d$. Kari–Seki [5] improve that general sufficient bound to $2p + q - d - \lfloor d/2 \rfloor$ for $p > q \geq 2d$. These hypotheses and conclusions differ from the alternating-prefix assertion proved here.

Simpson’s published Theorem 3.4 [6] concerns palindromic periodicities with half-periods h_1, h_2 and reflection offsets r_1, r_2 , and requires length at least $2h_1 + 2h_2 - D$, where $D = \gcd(2(r_2 - r_1), 2h_1, 2h_2)$; its conclusion is an ordinary period D . At the common alternating phase, $h_1 = p, h_2 = q$ and $D = 2d$, so that condition is $2p + 2q - 2d$, twice the length threshold in Theorem 1. Thus this close prior result does not directly give the short bound on w .

A bounded priority audit inspected these primary results, related proofs, bibliographies, equivalent formulations, and 53 recorded literature queries. It found no earlier exact theorem in that scope. This finding does not certify first discovery or universal novelty. The accompanying audit records its specific access and version gaps, including the unavailable Kari–Seki publisher full text and an unrecovered earlier conference version

on palindromes and local periodicity. The cited Kari–Seki comparison uses its dated author manuscript; the Czeizler–Kari–Seki comparison uses the actual 2010 journal-layout version, separately from the MFCS 2008 version.

The verification supplement preserves the submitted proof and programs, three independent mathematical audit families, the bounded priority record, complete finite-control outputs, and checksums. The families use signed constraints and a reflection lift, direct word-overlap checks, and definition/counterexample tests with canonical finite models. They include fixed letters, swapped letters, short truncations, divisibility cases, and below-threshold failures. These computations supplement the universal proof above; they do not replace it or certify priority. Public verification explicitly omits private source downloads and execution histories.

AI use and review status. AI tools were used extensively in solving, proof checking, literature research, drafting, computation, and independent adversarial agent reviews. This is an unrefereed research note. Those automated reviews are not independent external human peer review; no such human review is claimed.

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